Franco Huidobro

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Education

Tecnológico de Monterrey, Monterrey, Mexico — June 2025
MSc in Science of Engineering with Concentration in Power Electronics and Control Systems — GPA 4.0/4.0

- **Master's Thesis:** *A Controller for an LCI to Extend the Operating Range of Single-Phase Brushless Exciter Synchronous Motors.* Developed an advanced control strategy for load-commutated inverter (LCI) drives for SAG mills, introducing a novel flux-based load-synchronized pulse scheme and real-time inverter firing angle adjustment. Validated the system using hardware-in-the-loop with dSPACE hardware (HIL) simulations under realistic industrial milling profiles, successfully extending the motor's speed and load operating range.
  Published in the Tecnológico de Monterrey Institutional Repository (RITEC).

  Advisors: Dr. Gerardo Escobar, Dr. Renato Galluzzi, M. Rommel Villarreal.
  Funding: Research fully financed by AMI Automation.

- **Supplementary Analysis:** Developed a high-fidelity mathematical model to evaluate wireless power transfer (WPT) as an alternative excitation method for synchronous motors. Analyzed system behavior at low-speed/standstill conditions to propose efficiency improvements over traditional inductive coupling.

Tecnológico de Monterrey, Monterrey, Mexico — June 2023
BSc in Electromechanical Engineering — GPA 4.0/4.0

- **Minor:** Aeronautical Design & Manufacturing — GPA 4.0/4.0

- **Relevant Coursework:** Electric machines and electromechanical energy conversion, control and power electronics for electric machinery, industrial power systems, intelligent electrical networks.

Training and certification

English C1
French B2
MATLAB/Simulink and LabView courses.
Black Belt Lean Six Sigma Certification.
Train the Trainer Certification.
Green Belt Design for Six Sigma Certification.

Professional Experience

AMI Automation, Monterrey, Mexico — August 2023
R&D Power Electronic Control Firmware Developer.
Specialized in the full-lifecycle design of advanced control strategies for high-power thyristor and IGBT drives in heavy industries (steel, mining, metal processing).

- **R&D Project: Open-Winding Synchronous Cycloconverter (Steel Industry)**
  Scope: Designed a novel cycloconverter drive system for retrofitting tandem and reversible rolling mills.
  Theoretical Framework: Engineered a high-precision control architecture based on robust physical modeling. Derived motor dynamics via differential equations to implement comprehensive vector control loops and advanced state estimation algorithms (including Kalman filtering) for optimal flux and torque regulation and zero-sequence current suppression.
  Implementation & Deployment: Constructed high-fidelity plant models in MATLAB/Simulink for stability validation. Translated control laws into LabVIEW/NI FPGA for standalone execution.

- **R&D Project: Model Predictive Control (MPC) for Electric Arc Furnaces**
  Scope: Designed a high-speed rectifier control system to mitigate grid flickering and regulate DC current under chaotic load dynamics in steel process.
  Theoretical Framework: Engineered a model predictive control (MPC) algorithm to predict the current dynamic response and reject flickering.
  Implementation & Deployment: Validated controller robustness against a chaotic load via Simulink simulations.

- **R&D Project: Supersynchronous Induction Drive**
  Scope: Developed the firmware for a wound-rotor induction motor drive aimed at high-efficiency supersynchronous speed operation.
  Theoretical Framework: Formulated an optimal operating point (OOP) tracking algorithm to maximize efficiency across variable load torques. Implemented complex regulation loops for reactive power (VARs) and rotor current injection. The drive designed is sensorless.
  Implementation & Deployment: Constructed high-fidelity plant models in MATLAB/Simulink for stability validation. Translated control laws into LabVIEW/NI FPGA for standalone execution.

- **Field Project: Quadra-Torque Drive System (Minera Peñasquito)**
  Scope: Led the end-to-end commissioning of four dual-synchronous motor ball mill drives, ensuring seamless integration with existing plant infrastructure.
  Leadership & Strategy: Led the installation workflow, supervising technician teams to reduce downtime. Served as the primary technical liaison for troubleshooting during the critical startup phase.
  Knowledge Transfer: Designed and delivered an advanced technical training curriculum for 70 site engineers, covering drive theory, fault analysis, and operational safety.

- **R&D Project: Power Quality & Energy Storage (Resonac Graphite)**
  Scope: Analyzed grid disturbances during silicon crystal growth and engineered a supercapacitor-based solution to protect critical loads.
  Theoretical Framework: Modeled the full rectifier-inverter topology and grid interaction dynamics. Calculated optimal supercapacitor sizing and designed a soft-precharge control logic to limit inrush currents.
  Implementation & Deployment: Constructed high-fidelity plant models in MATLAB/Simulink for stability validation. Translated control laws into LabVIEW/NI FPGA for standalone execution.

Tecnológico de Monterrey, Monterrey, Mexico August 2023 – June 2025
Graduate Teaching Assistant & Laboratory Instructor

- **Power Electronics Laboratory:** Instructed laboratory sessions on DC-DC and DC-AC converters, supervising students in circuit analysis, hardware debugging, and safety protocols.
- **Program Coordination** (Dr. Esmeralda Uribe Lam): Collaborated with the program director to redesign engineering curricula presented to the board. Evaluated performance in advanced aeronautical and materials engineering coursework.

Tecnológico de Monterrey, Monterrey, Mexico November 2022 - August 2023
Research Assistant

Embedded Control: Designed a closed-loop AC motor speed control system using a Microchip dsPIC and C language, integrating a three-phase inverter with encoder feedback.
Deployed the prototype as an active educational module and presented the technical design at "Expo-Ingenierías" (University Science Showcase).

Tecnológico de Monterrey/Lean Methods Group, Monterrey, Mexico August 2021 - July 2022
Lean Six Sigma Advisor

Process Optimization: Co-led manufacturing projects (Ceramic/Tantalum capacitors) achieving $660k USD/year in savings using Minitab for statistical analysis (ANOVA, DOE, Gage R&R).
Certifications: Earned Lean Six Sigma Black Belt and Design for Six Sigma (DFSS) Green Belt.

Leadership & Extracurricular Activities

Additive Manufacturing August 2022 - December 2022
Co-founder and president of student group MATec

Founded a student group focused on 3D printing solutions. Led a project to redesign and print plastic components for a local manufacturer, solving a critical supply chain shortage and saving the company from costly machinery replacements.

Sustainability at colleges January 2019 - December 2022
Co-coordinator of student group Proyecto 25

Organized community activities focused on environmental conservation, reforestation, and sustainable nutrition. Promoted eco-friendly practices and awareness within the student body and local neighborhoods.

Arts promotion 2016-2017
Pianist in the International Art Festival 2016 and 2017 editions.

Selected to perform as a pianist in two consecutive editions of the festival, demonstrating discipline and a long-term commitment to the arts.

Honors, Awards & Affiliations

Professional Memberships

Member, IEEE (Institute of Electrical and Electronics Engineers).

National Award CENEVAL 2023

Top 1.5% Nationwide: Awarded by the National Evaluation Center for Higher Education (Mexico) for achieving outstanding performance across all areas of the General Exit Examination for Bachelor's Degrees (EGEL).

Honorable Mention of Excellence (Summa Cum Laude) 2023

Top 2.5% of Class: Highest academic distinction awarded by Tecnológico de Monterrey to graduates with exceptional GPA and academic integrity.

Outstanding Trajectory of the School of Engineering and Sciences. 2023

Recognized for maintaining a balance of excellence between academic research, industrial internships, and extracurricular leadership during the undergraduate degree.

Leadership in Professional Development LIFE Award 2023

Distinguished for demonstrating exceptional leadership and problem-solving capabilities during professional internships and industrial placements.

Interests & Skills

Research Interests

- **Advanced Electric Drives:** High-power cycloconverter for high-power applications, multiphase machines, and fault-tolerant control.
- **Nonlinear Control:** Model Predictive Control (MPC), sliding mode control and state estimation.
- **Power Systems:** Grid-tied power electronics, renewable energy integration, and energy storage management.

Technical Skills

- **Modeling & Simulation:** MATLAB, Simulink (Simscape Electrical), Stateflow, Mathematical Modeling (Differential Equations), SPICE.
- **Embedded Control & FPGA:** LabVIEW Real-Time, NI FPGA Module, NI cRIO, Microchip dsPIC, C/C++ Programming for Embedded Systems.
- **Hardware & Testing:** Hardware-in-the-Loop (HIL), Rapid Control Prototyping (RCP), Power Analyzers, **Electronic Instrumentation**, **Industrial Commissioning**.
- **Analysis & Quality:** Minitab (DOE, ANOVA, Hypothesis Testing), Lean Six Sigma Black Belt, Design for Six Sigma (DFSS).